Chpater2 Thermal & Wave Motion

Lesson1 Thermal Physics, Expansion & Heat

1. For each of the following temperatures, find the equivalent temperature on the indicated scale: **(a)** $2273.15°\text{C}$ on the Fahrenheit scale, (b) $98.6°\text{F}$ on the Celsius scale, and (c) $100\ \text{K}$ on the Fahrenheit scale.

2. A nurse measures the temperature of a patient to be $41.5°\text{C}$. **(a)** What is this temperature on the Fahrenheit scale? **(b)** Do you think the patient is seriously ill? Explain.

3. Death Valley holds the record for the highest recorded temperature in the United States. On July 10, 1913, at a place called Furnace Creek Ranch, the temperature rose to $134°\text{F}$. The lowest U.S. temperature ever recorded occurred at Prospect Creek Camp in Alaska on January 23, 1971, when the temperature plummeted to $279.8°\ \text{F}$. **(a)** Convert these temperatures to the Celsius scale. **(b)** Convert the Celsius temperatures to Kelvin.

4. The boiling point of liquid hydrogen is $20.3\ \text{K}$ at atmospheric pressure. What is this temperature on **(a)** the Celsius scale and **(b)** the Fahrenheit scale?

5. Show that if the temperature on the Celsius scale changes by ΔT_{C}, the Fahrenheit temperature changes by $\Delta T_F = \frac{9}{5}\Delta T_C$.

6. The temperature difference between the inside and the outside of a home on a cold winter day is $57.0°\text{F}$. Express this difference on **(a)** the Celsius scale and **(b)** the Kelvin scale.

7. The pressure in a constant-volume gas thermometer is $0.700\ \text{atm}$ at $100°\text{C}$ and $0.512\ \text{atm}$ at $0°\text{C}$. **(a)** What is the temperature when the pressure is $0.040\ 0\ \text{atm}$? **(b)** What is the pressure at $450°\text{C}$?

8. The highest recorded waterfall in the world is found at Angel Falls in Venezuela. Its longest single waterfall has a height of $807\ \text{m}$. If water at the top of the falls is at $15.0°\text{C}$, what is the maximum temperature of the water at the bottom of the falls? Assume all the kinetic energy of the water as it reaches the bottom goes into raising the water's temperature.

9. The temperature of a silver bar rises by $10.0°\text{C}$ when it absorbs $1.23\ \text{kJ}$ of energy by heat. The mass of the bar is $525\ \text{g}$. Determine the specific heat of silver from these data.

10. Lake Erie contains roughly $4.00 \times 10^{11}\ \text{m}^3$ of water. **(a)** How much energy is required to raise the temperature of that volume of water from $11.0°\text{C}$ to $12.0°\text{C}$? **(b)** How many years would it take to supply this amount of energy by using the $1\ 000\text{-MW}$ exhaust energy of an electric power plant?

11. A 1.50-kg iron horseshoe initially at $600°\text{C}$ is dropped into a bucket containing $20.0\ \text{kg}$ of water at $25.0°\text{C}$. What is the final temperature of the water– horseshoe system? Ignore the heat capacity of the container and assume a negligible amount of water boils away.

12. A student drops two metallic objects into a 120-g steel container holding $150\ \text{g}$ of water at $25°\text{C}$. One object is a 200-g cube of copper that is initially at $85°\text{C}$, and the other is a chunk of aluminum that is initially at

$5.0°C$. To the surprise of the student, the water reaches a final temperature of $25°C$, precisely where it started. What is the mass of the aluminum chunk?

13. Equal 0.400-kg masses of lead and tin at $60.0°C$ are placed in 1.00 kg of water at $20.0°C$. **(a)** What is the equilibrium temperature of the system? **(b)** If an alloy is half lead and half tin by mass, what specific heat would you anticipate for the alloy?

14. An unknown substance has a mass of 0.125 kg and an initial temperature of $95.0°C$. The substance is then dropped into a calorimeter made of aluminum containing 0.285 kg of water initially at $25.0°C$. The mass of the aluminum container is 0.150 kg, and the temperature of the calorimeter increases to a final equilibrium temperature of $32.0°C$. Assuming no thermal energy is transferred to the environment, calculate the specific heat of the unknown substance.

15. A 75-g ice cube at $0°C$ is placed in 825 g of water at $25°C$. What is the final temperature of the mixture?

16. A 50-g ice cube at $0°C$ is heated until 45 g has become water at $100°C$ and 5.0 g has become steam at $100°C$. How much energy was added to accomplish the transformation?

17. A 100-g cube of ice at $0°C$ is dropped into 1.0 kg of water that was originally at $80°C$. What is the final temperature of the water after the ice has melted?

18. How much energy is required to change a 40-g ice cube from ice at $210°C$ to steam at $110°C$?

19. A 40-g block of ice is cooled to $278°C$ and is then added to 560 g of water in an 80-g copper calorimeter at a temperature of $25°C$. Determine the final temperature of the system consisting of the ice, water, and calorimeter. (If not all the ice melts, determine how much ice is left.) Remember that the ice must first warm to $0°C$, melt, and then continue warming as water. (The specific heat of ice is 0.500 cal/g ? °C 5 2 090 J/kg ? °C.)

20. Steam at $100°C$ is added to ice at $0°C$. **(a)** Find the amount of ice melted and the final temperature when the mass of steam is 10 g and the mass of ice is 50 g. **(b)** Repeat with steam of mass 1.0 g and ice of mass 50 g.

21. A 3.00-g lead bullet at $30.0°C$ is fired at a speed of 2.40×10^2 m/s into a large, fixed block of ice at $0°C$, in which it becomes embedded. **(a)** Describe the energy transformations that occur as the bullet is cooled. What is the final temperature of the bullet? **(b)** What quantity of ice melts?

22. A 40-g ice cube floats in 200 g of water in a 100-g copper cup; all are at a temperature of $0°C$. A piece of lead at $98°C$ is dropped into the cup, and the final equilibrium temperature is $12°C$. What is the mass of the lead?

23. A 200-g block of copper at a temperature of $90°C$ is dropped into 400 g of water at $27°C$. The water is contained in a 300-g glass container. What is the final temperature of the mixture?

24. Liquid helium has a very low boiling point, $4.2\ \text{K}$, as well as a very low latent heat of vaporization, $2.00 \times 10^4\ \text{J/kg}$. If energy is transferred to a container of liquid helium at the boiling point from an immersed electric heater at a rate of $10.0\ \text{W}$, how long does it take to boil away $2.00\ \text{kg}$ of the liquid?

25. One mole of oxygen gas is at a pressure of $6.00\ \text{atm}$ and a temperature of $27.0°\text{C}$. **(a)** If the gas is heated at constant volume until the pressure triples, what is the final temperature? **(b)** If the gas is heated so that both the pressure and volume are doubled, what is the final temperature?

26. (a) An ideal gas occupies a volume of $1.0\ \text{cm}^3$ at $20°\text{C}$ and atmospheric pressure. Determine the number of molecules of gas in the container. (b) If the pressure of the 1.0-cm3 volume is reduced to $1.0 \times 10^{-11}\ \text{Pa}$ (an extremely good vacuum) while the temperature remains constant, how many moles of gas remain in the container?

27. An automobile tire is inflated with air originally at $10.0°\text{C}$ and normal atmospheric pressure. During the process, the air is compressed to 28.0% of its original volume and the temperature is increased to $40.0°\text{C}$. **(a)** What is the tire pressure in pascals? **(b)** After the car is driven at high speed, the tire's air temperature rises to $85.0°\text{C}$ and the tire's interior volume increases by 2.00%. What is the new tire pressure (absolute) in pascals?

28. Gas is confined in a tank at a pressure of $11.0\ \text{atm}$ and a temperature of $25.0°\text{C}$. If two-thirds of the gas is withdrawn and the temperature is raised to $75.0°\text{C}$, what is the new pressure of the gas remaining in the tank?

29. Gas is contained in an 8.00-L vessel at a temperature of $20.0°\text{C}$ and a pressure of $9.00\ \text{atm}$. **(a)** Determine the number of moles of gas in the vessel. **(b)** How many molecules are in the vessel?

30. An air bubble has a volume of $1.50\ \text{cm}^3$ when it is released by a submarine $100\ \text{m}$ below the surface of a lake. What is the volume of the bubble when it reaches the surface? Assume the temperature and the number of air molecules in the bubble remain constant during its ascent.